BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

SQL Queries

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why open a file before reading it?

2. What does CPU cache hold?
   A) Only saved personal files
   B) Often-used data and instructions
   C) Only website passwords
   D) Only the next address

3. "A register is a large external drive used for backups." Correct this. Explain why.

4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

Today's objectives

Explain the core ideas of sql queries using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

SELECT
Identifies which field or fields a query retrieves.

FROM
Names the table containing the records.

WHERE
Filters rows using a condition.

SQL query
A database query retrieves only the requested information from a table.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Field and record
A SQL table stores each pupil as a row/record; Name and Mark are columns/fields.

SELECT
Names the field(s) to return in the result, such as Name or ID.

FROM
Names the table to search; our shared example is Pupils.

WHERE
Filters rows. Comparing the numeric Mark to 50 returns only pupils meeting the condition.

I Do - Worked model

Using Pupils(ID, Name, Mark): 101 Ada 72; 102 Ben 43; 103 Mia 81. Write SQL to return the names of pupils with Mark >= 50, and state the result.

- SELECT Name: ask for the Name field, not the complete record.
- FROM Pupils: identify the table holding these rows.
- WHERE Mark >= 50: Ada (72) and Mia (81) match; Ben (43) does not.
- SELECT Name FROM Pupils WHERE Mark >= 50; returns Ada and Mia, in no guaranteed order unless an ordering rule is specified.

We Do - Guided practice

1. Using the same Pupils table, write SQL to return the ID of pupils with Mark < 50. Identify the result. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write an SQL query returning Name where Class = 11.

```
SELECT Name FROM Pupils ...
```

You Do - Independent exam questions

Q2. Write SQL to return Name from Pupils where Mark > 70. [3 marks]

Q3. For the shared table, what value is returned by SELECT Name FROM Pupils WHERE ID = 103;? [2 marks]

Q4. Explain how SELECT and WHERE perform different jobs in SELECT Name FROM Pupils WHERE Mark < 50; [3 marks]

Exit Ticket - True or False?

SQL WHERE selects the table name.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Opening establishes a handle or connection through which the program accesses the file.

2. B - Often-used data and instructions

3. A small, fast storage location inside the CPU. It holds data or addresses in use.

4. The accumulator holds the result, 18. The PC holds the next instruction's address, 240.

I Do - Explain the model

SELECT Name: ask for the Name field, not the complete record.

FROM Pupils: identify the table holding these rows.

WHERE Mark >= 50: Ada (72) and Mia (81) match; Ben (43) does not.

SELECT Name FROM Pupils WHERE Mark >= 50; returns Ada and Mia, in no guaranteed order unless an ordering rule is specified.

We Do - Model answers

1. SELECT ID FROM Pupils WHERE Mark < 50; returns 102. SELECT/FROM/WHERE [3] and correct resulting row [1].

Code First - Answer

SELECT Name FROM Pupils WHERE Class = 11;
Expected output: names from matching rows (table supplied)

You Do - Self-mark

Q2: SELECT Name [1] FROM Pupils [1] WHERE Mark > 70 [1]; result Ada and Mia.

Q3: Mia [1]; because the row with ID 103 matches the WHERE condition [1].

Q4: SELECT Name chooses the field to output [1]; WHERE Mark < 50 filters qualifying rows [1]; in this dataset only Ben matches [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.